Review of the first class

- Game theory studies decision-making in environments with *strategic interactions*
- A rational player should adopt a *dominant strategy* if any
- A rational player should avoid using *dominated strategies*

The Running Example: Software Launch Game

- Many games have no dominant or dominated strategies
- Then how to make a prediction?

		Your Rival: Launch	Your Rival: Not
You	Launch	-20, -20	30, -10
You	Not	-10, 30	0, 0

Nash Equilibrium

- **Definition**: A *Nash Equilibrium (NE)* is a profile of strategies (one strategy for each player) in which *no* player can improve her payoff by *unilaterally* changing her strategy

  - A NE is a "*stable situation*" where each player is doing their best given what the other players are doing
  - In other words, no player has an incentive to leave NE *individually* if the game is played again, or there is no *ex-post individual* regret

Example

- The prediction made by dominant/dominated strategies must be a NE
- Recall this advertising competition game:

		Philip Morris	
		H Ad	L Ad
Reynolds	H Ad	20, 20	50, 10
	L Ad	10, 50	40, 40

- No firm has an *individual* incentive to leave the NE (H Ad, H Ad)
- The better outcome (L Ad, L Ad) is, unfortunately, *not* a NE, as each firm would have an *individual* incentive to switch to H Ad

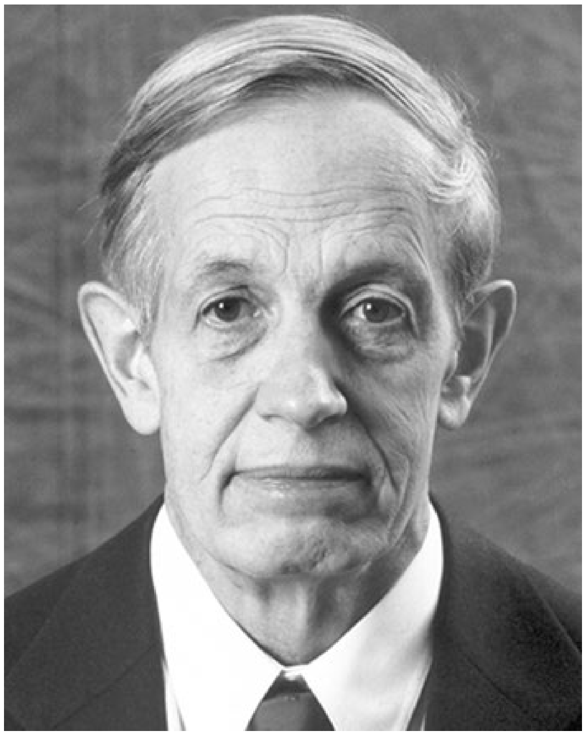
Example

- What's the NE in this running example? Let's check and verify …
  - There are TWO Nash equilibria
  - Which one to pick? Will come back to this issue when we discuss coordination games

		Your Rival: Launch	Your Rival: Not
You	Launch	-20, -20	30, -10
You	Not	-10, 30	0, 0

A Beautiful Mind

Want to know more about John Nash? Watch the movie or see Wiki



John F. Nash (1928-2015)
Noble Laureate in Economics, 1994

Unfortunately, the director Ron Howard's understanding of NE was *wrong*

Example

- It is too time consuming to check and verify …

Player II

Player I		L	C	R
	T	9, 5	3, 6	1, 7
	M	1, 5	4, 3	2, 2
	B	2, 7	3, 6	3, 8

Best-Response Method

- Instead of "check and verify," a more efficient way to find NE is the *best-response method*
  - ***Step 1***: Find each player's *best response*---her best choice for *each* possible move by her rivals
  - ***Step 2***: The *intersection* of all players' best responses is a NE (i.e., the cell in which all players are at a best response)

Best-Response Method

- Find NE in the following game:

		Player II		
		L	C	R
Player I	T	9, 5	3, 6	1, 7
	M	1, 5	4, 3	2, 2
	B	2, 7	3, 6	3, 8

(B,R) is the NE

- The best-response method is a way to help find NE, but not necessarily the way how players think through a game
- It, however, suggests a *learning foundation* for NE: players may not know the concept of NE, but if they can adjust their choices over time by *trial-and-error*, the outcome oftentimes converges to NE.

Summary of the theory so far

- Game theory uses Nash equilibrium or "stable situation" to make predictions
  - No individual incentive to deviate; no *ex post* regret; often has a learning foundation

- Possible approaches to find NE
  - Dominant or dominated strategy (if exists)
  - Best-response method
  - Guess and verify
    (not very efficient, but sometimes useful)

The Prisoner’s Dilemma

- Two suspects are arrested for a minor crime (e.g., carrying unregistered guns) but are suspected of a major crime (e.g., bank robbery)
- Each of them can choose whether to confess to the major crime or remain silent
  - If neither confesses, each will serve one year in jail
  - If both confess, each will serve six years in jail
  - If one confesses and the other keeps silent, the confessor will be released immediately and the other will be sentenced to nine years in jail

The Prisoner's Dilemma …

		Prisoner 2	
		Silent	Confess
Prisoner 1	Silent	-1, -1	-9, $\underline{0}$
	Confess	$\underline{0}$, -9	$\underline{-6}$, $\underline{-6}$

Note that "Confess" is also a dominant strategy for each player

- Individual rationality leads to the NE (Confess, Confess), but (Silent, Silent) is a collectively better outcome---***Prisoner's Dilemma***
- In one-shot game without third-party intervention, not easy to settle on (Silent, Silent)

The Prisoner’s Dilemma …

- This simple game illustrates an important insight:

  individually “right” actions can lead to collectively “wrong” outcome

  - Similar flavor as the advertising competition game before

- Nash Equilibrium does NOT guarantee cooperation

- Against the classic view of “invisible hand” by Adam Smith

Prisoner's Dilemma in Real Life

- Advertising/price competition
- The tragedy of the commons (Garrett Hardin, 1968)
  - People tend to overuse public resources: overgrazing, overfishing, pollution, global warming, etc.
- Trade wars; currency wars
- Arm races
- Relative performance evaluation (e.g., grading curves)
- Doping in sports
- Excessive competition in child education
- New technology races (e.g., AI, stable coins) and over-deregulation


THE WHITE HOUSE

Winning the Race

AMERICA'S AI ACTION PLAN

JULY 2025

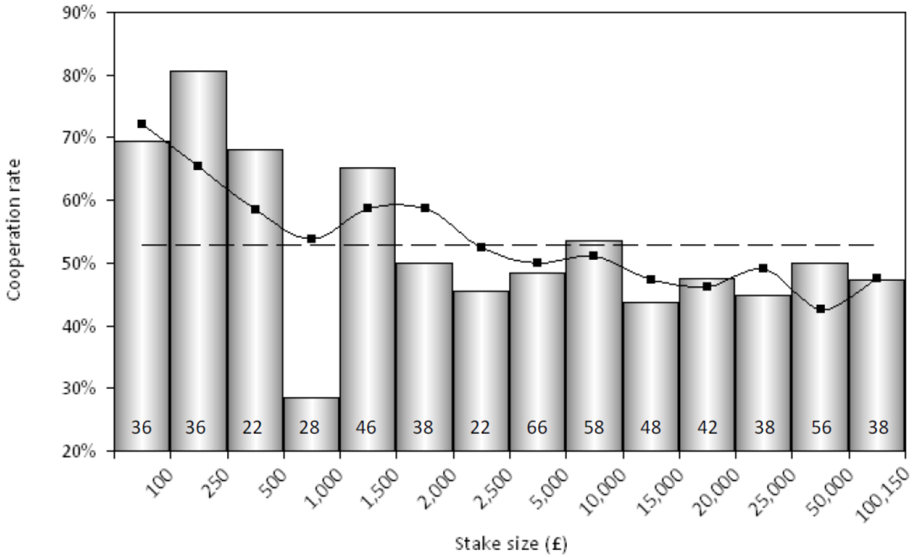
"Today, a new frontier of scientific discovery lies before us, defined by transformative technologies such as artificial intelligence... Breakthroughs in these fields have the potential to reshape the global balance of power, spark entirely new industries, and revolutionize the way we live and work. As our global competitors race to exploit these technologies, it is a national security imperative for the United States to achieve and maintain unquestioned and unchallenged global technological dominance. To secure our future, we must harness the full power of American innovation."

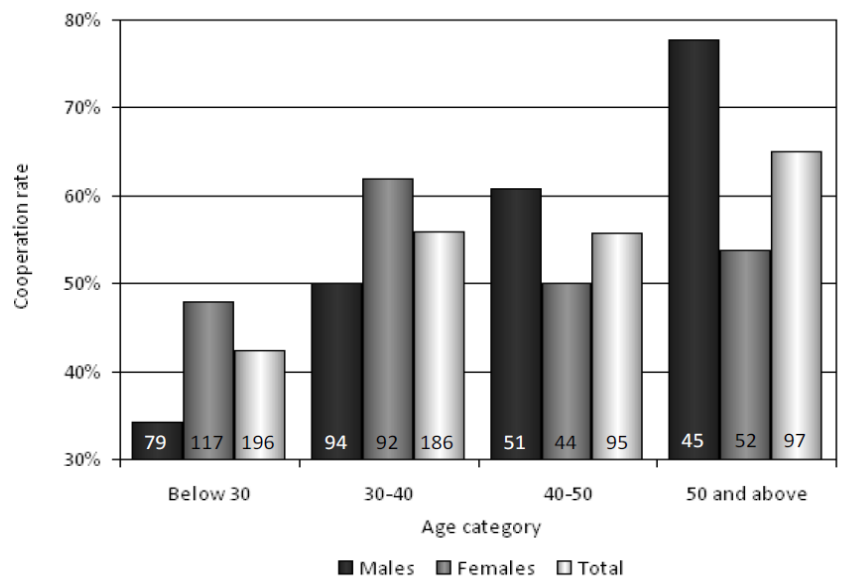
Donald J. Trump

45th and 47th President of the United States

Escape Prisoner’s Dilemma

- Possible ways to escape the Prisoner’s Dilemma?
  - Sequential moves?
  - Communication? (https://www.youtube.com/watch?v=TKaYRH6E36U)
  - Contracting/agreement:
    - WTO, G7/G20, Paris Agreement, World Anti-Doping Agency, etc.
  - Social norm; long-term relationship (or even religion)





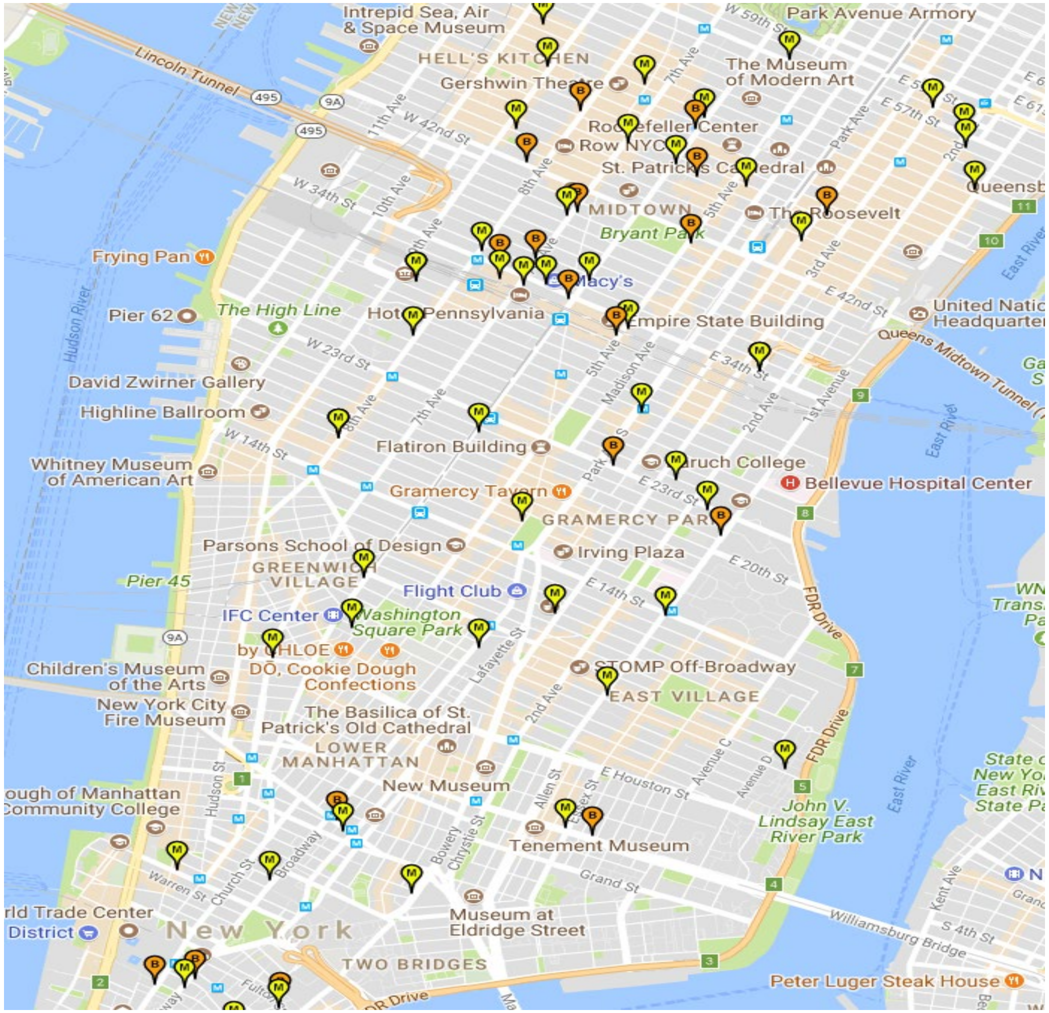
Data from the Golden Ball game

Location Choice Game

- Two ice cream vendors choose where to locate on a beach
- Tourists are evenly distributed on the beach
- Suppose price is fixed, and tourists prefer the closer vendor
- Where will the two vendors choose to locate?

- Impossible to draw a matrix, but we can guess and verify …
- Key insight: shifting toward the competitor to "steal" tourists in the middle

- Homework: what about if there are three vendors?



MacDonald and Burger King in NYC

(http://www.fastfoodmaps.com)

Location Game

- Policy standing choice in campaigns
- Product design as "location" choice

Smartphone commoditization?

Summary

- Game theory studies decision-making when *strategic interaction* is present
  - *Strategic thinking* is crucial; put yourself into other players' shoes

- We use *Nash Equilibrium (NE)* to make predictions
  - Stable situation with no *individual* deviation incentive
  - Method 1: Dominant or dominated strategy (if any)
  - Method 2: Best-response method
  - Method 3: Guess and verify

- Lessons from important games discussed so far:
  - *Guessing 2/3 of the average*: other players' sophistication; learning
  - *Prisoner's dilemma*: individual rationality vs collective rationality
  - *Location choice game*: shifting toward the competitor to attract consumers/voters in the middle